What Survives a Delta Hedge?

Curvature, Jump Exposure, and the Maturity Clock in SPX Options

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Abstract

Large Delta-hedging errors near expiration are primarily a curvature phenomenon, not a discontinuity phenomenon. As expiry approaches, the hedged option sheds volatility-level exposure while its curvature diverges, so the natural null for the residual is not zero but the Boyle-Emanuel discrete-hedging floor. This paper separates that floor from whatever discontinuities add on top of it, using 1,243,969 intraday observations of SPXW zero-day-to-expiry options on 1,379 trading dates from September 2016 to May 2024. Realized spot variation over each contract's own life is split into continuous and discontinuous components by Lee-Mykland detection and scaled by the contract's own curvature. Continuous variation accounts for the held-Delta residual with a loading of 0.939 and a date-clustered standard error of 0.079, consistent with the unit benchmark. Identified discontinuous variation adds nothing beyond its contribution to total quadratic variation: the contrast $\beta_J - \beta_C$ is -0.106 with a p-value of 0.411. Because jump detection selects unusually large returns, that contrast is read against a pure-diffusion placebo in which no jump and no premium exist by construction; across 340 replications the placebo contrast has a mean of -0.054 and a standard deviation of 0.357, and 48 percent of replications fall below the estimate. A parity proposition predicts that the mechanical jump-convexity component is shared by matched calls and puts, and on 565,573 matched pairs the mean residual difference is 0.055 basis points against a mean absolute residual of 5.608. Identification rests on 39 jump dates rather than on the row count, and the design has eighty percent power only against an excess near one times the continuous loading, so the result is a well-identified null rather than a sharp bound.

Keywords: SPX options; Delta hedging; jump risk; replication residual; Gamma; 0DTE; short maturity.
JEL: G12, G13, G17.

1 Introduction

A Delta hedge removes the first derivative of option value with respect to spot at the instant the hedge ratio is chosen. It does not remove curvature, stochastic volatility, non-spot state variables, discontinuities, or the wedge between physical and pricing-measure risk. The object studied here is the return left after that local slope has been neutralized, and the central question is how much of it identified discontinuities explain once ordinary curvature has been accounted for.

The question is sharpest at the shortest maturities, which is also where the market now is: contracts with a week or less to expiry carry the bulk of SPX option volume (Bandi et al., 2026). As

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expiry approaches, Vega vanishes at rate $\sqrt{T}$ while Gamma diverges at rate $T^{-1/2}$, so the hedged position becomes almost entirely a position in the curvature of the spot path. That is what makes the ultra-short segment the natural place to look for jump effects, and it is also what makes them hard to see, because curvature is the channel through which a jump would have to act.

The paper's claim is this. Large hedging errors in ultra-short options are curvature phenomena, and once continuous quadratic variation is put on its theoretical Gamma scale, detected discontinuities carry no additional positive loading. In the intraday SPXW zero-day-to-expiry panel, curvature-scaled continuous variation accounts for the held-Delta residual with a loading statistically indistinguishable from the discrete-diffusion value of one. Discontinuous variation, identified by three distinct high-frequency estimators, adds nothing above it: the incremental jump coefficient is slightly negative and never distinguishable from the continuous coefficient in the preferred specification. Everything that follows exists to make that null credible rather than merely unrejected.

That finding is only meaningful if the design defeats a specific contamination. Jump detection selects unusually large returns, so a jump variable can load on hedging error even when the price path is a pure diffusion: for a given quadratic variation, a path whose variation is concentrated in one increment produces a terminal move of $\sqrt{QV}$ where a Gaussian path produces $\sqrt{2QV/\pi}$ on average. A jump dummy is therefore mechanically correlated with curvature-driven hedging error under the null of no jumps at all. This paper handles the problem in two ways. First, quadratic variation is decomposed additively rather than dummied, so the continuous and discontinuous components enter with the same curvature scaling and the economically relevant object is the contrast between their coefficients. Second, the contrast is calibrated by simulation: Gaussian paths with a jump-robust variance profile and the empirical intraday periodicity are generated, options are priced with the same variance that generates the path, and the full detection and estimation pipeline is rerun. Under that null there is no jump and no premium by construction, and the recovered contrast measures detection selection alone. The estimated contrast lies inside the body of that distribution.

Three further results follow. The paper's parity proposition states that for matched European calls and puts hedged with parity-consistent Deltas, the jump component of the hedged residual is one and the same object. The intraday panel puts calls and puts on an identical strike grid and therefore tests the proposition directly rather than by proxy. Across 565,573 matched pairs the residual difference has a mean of 0.055 basis points and a mean absolute value of 1.076 basis points, against a mean absolute residual of 5.608 basis points; what remains tracks the measured parity violations in quoted mid prices and vendor Deltas rather than any convexity channel. Second, the maturity clock inside the session is informative on its own terms: mean absolute residual rises monotonically from 2.00 basis points at thirty minutes to expiry to 9.16 basis points at six hours, and the continuous loading rises with horizon while the jump loading does not. Third, the parity result carries a restriction beyond this sample: any call-put asymmetry in jump-related hedging returns, in this panel or another, cannot be produced by payoff convexity, so the explanation must lie in pricing, demand, or measurement.

The paper contributes in four ways. It executes a modern intraday test of the continuous-versus-discontinuous decomposition on a panel that contains genuine jump events, including the February 2018 volatility episode, the March 2020 crash, and the December 2017 intraday reversal. It supplies a placebo calibration that makes a null on jump excess interpretable rather than merely unsignificant. It converts the parity proposition from a theoretical remark into a measured quantity. And it delimits the claim: the design identifies mechanical jump exposure and jump-associated excess residual, and it does not identify a jump-risk premium, a distinction maintained throughout.

1.1 Related literature

The paper sits at the intersection of three strands. The delta-hedged gains literature, beginning with Bakshi and Kapadia (2003), uses hedged option returns to isolate the volatility risk premium; Broadie et al. (2007) and Pan (2002) estimate jump risk premia structurally, and Bollerslev and Todorov (2011) separate fear of jumps from diffusive variance compensation. The framework below differs in bookkeeping rather than in mechanism: it groups everything a local hedge cannot remove into four channels and asks which channels observable jump statistics can reach. Second, the discrete and imperfect hedging literature characterizes the mechanical part of the residual: Boyle and Emanuel (1980) derive the distribution of discretely rebalanced hedge errors under pure diffusion, and Hull and White (2017) show that a minimum-variance Delta accounting for the empirical spot-volatility relation removes part of the practitioner Delta's error. Both objects appear below as quantitative benchmarks that the data are asked to match, not as citations. Third, the short-maturity literature documents that risk near expiration is economically distinct: Andersen et al. (2017) extract short-term risk premia from weekly options, Bandi et al. (2026) model ultra-short volatility surfaces, and Božović (2025) reports intraday jump compensation in 0DTE contracts at five-minute frequency. The jump-identification machinery follows Lee and Mykland (2008), Barndorff-Nielsen and Shephard (2004), and the truncation approach surveyed in Jiang and Oomen (2008). The pricing side of that literature descends from Merton (1976) and the affine jump-diffusion transform methods of Duffie et al. (2000), and the empirical case that discontinuity risk carries compensation in index options has been made from several directions: Santa-Clara and Yan (2010) from the crash-adjusted equity premium, Yan (2011) from the slope of the implied-volatility smile, Kelly and Jiang (2014) from tail-risk exposure, Bégin et al. (2020) from joint equity and option evidence, and Chen et al. (2023) from a decomposition of straddle prices around anticipated jumps. The result below does not contradict that literature. It restricts where the compensation can be found: not in the held-Delta residual of ultra-short contracts, once curvature is scaled by its own theoretical constant. For scheduled discontinuities, Zhong (2026) shows that identification requires separating the no-event surface from event-spanning quotes, a restriction respected in Section 6.

Relative to Božović (2025), which reports intraday jump compensation in 0DTE contracts, the design here differs in the object measured and in the null imposed. The dependent variable is the held-Delta residual over the contract's own remaining life rather than a rebalanced return, the jump variable is a curvature-scaled variance component rather than an indicator, and the benchmark is a simulated jump-free distribution rather than zero. These differences are enough to change the sign of what a jump variable appears to explain, which is the methodological point of Section 4.

2 Theory and testable predictions

2.1 A state-based representation of hedged P&L

Let the pricing state be $X_t = (S_t, q_t)$, where S_t is the underlying index and q_t collects every additional state variable relevant for the option price. Under the physical measure, allow X_t to contain both continuous shocks and discontinuities. Using $K(X_t)$ for the mean jump compensator, a compact representation is

$$dX_t = [\mu(X_t) - K(X_t)]\,dt + \Sigma(X_t)\,dW_t + dJ_t, \tag{1}$$

where $\Sigma\,dW_t$ denotes the vector of diffusion innovations and dJ_t the vector of jumps. The formulation is intentionally broad. In a modern SPX application, q_t can be read as including volatility factors, coordinates of the volatility surface, liquidity or intermediary conditions, and other states that move option values without being identical to the spot index.

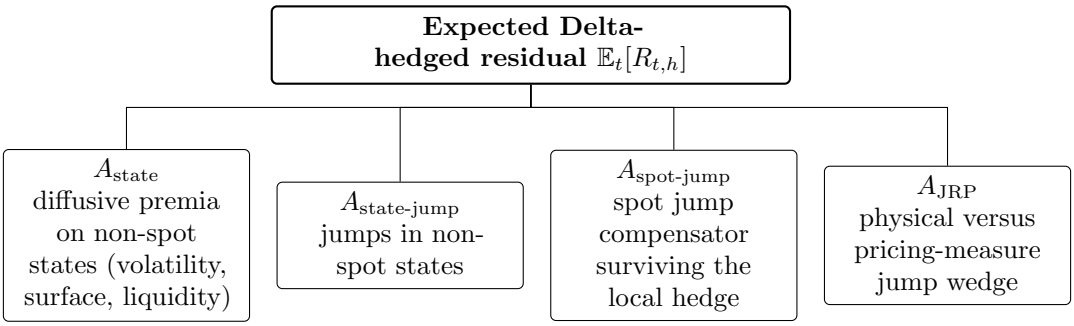


Figure 1: Four expected-return channels that can remain after local Delta neutralization.

For an option with value C_t, consider a self-financing position that is long the option and short its instantaneous spot Delta. Over a horizon h, define the financing-adjusted residual as

$$R_{t,h} = (C_{t+h} - C_t) - \int_t^{t+h} \Delta_u \, dS_u - \int_t^{t+h} r_u \, (C_u - \Delta_u S_u) \, du. \tag{2}$$

In a frictionless complete diffusion setting with continuous trading, this object collapses toward zero under perfect replication. Once jumps and additional priced states are admitted, Delta neutralization removes only the first-order continuous spot exposure. The remaining expected return can be grouped into economically distinct channels. Writing the decomposition schematically rather than tying it to one specific state parameterization gives

$$\mathbb{E}_t[R_{t,h}] = A_{\text{state}} + A_{\text{state-jump}} + A_{\text{spot-jump}} + A_{\text{JRP}}, \tag{3}$$

as illustrated in Figure 1, where A_{state} is compensation for diffusion risk in non-spot states, $A_{\text{state-jump}}$ captures jumps in those states, $A_{\text{spot-jump}}$ is the expected jump component of the underlying that survives the local hedge, and A_{JRP} is the difference between physical and risk-neutral jump behavior. The fuller stochastic expression also contains mean-zero diffusion innovations in the auxiliary states; those innovations disappear from the short-horizon conditional expectation but remain part of realized hedging P&L.

2.2 Maturity composition of the residual

The channels in Equation (3) do not carry equal weight at every maturity. The key short-maturity distinction is not "jumps versus everything else." It is spot-path nonlinearity versus smooth state sensitivity. The following results, proved in Appendix B, make that distinction explicit and delimit what convexity can and cannot explain.

Proposition 1 (Curvature concentration near expiration)**.** *Fix spot S, volatility $\sigma > 0$, and rate r, and consider Black-Scholes calls and puts with maturity T and strikes on the local-moneyness family $K = Se^{\kappa\sigma\sqrt{T}}$ for fixed $\kappa \in \mathbb{R}$, each hedged with its instantaneous Delta. As $T \to 0$, $d_1 \to -\kappa$, and: (i) Vega satisfies $\partial C/\partial\sigma = S\varphi(d_1)\sqrt{T} = O(\sqrt{T})$ and therefore vanishes, uniformly for κ in compact sets; (ii) Gamma satisfies $\Gamma = \varphi(d_1)/(S\sigma\sqrt{T}) = O(T^{-1/2})$ and therefore diverges, uniformly for κ in compact sets; (iii) for a small spot return $u = \Delta S/S$ while the hedge is held fixed, the leading hedge residual is $\frac{1}{2}\Gamma S^2 u^2 + o(u^2)$ for calls and puts alike, whether u is generated by a continuous diffusion increment or by a small jump; (iv) for a fixed nonzero log jump J, the exact hedge residuals of the call and of the put converge to the same strictly positive limit*

$$S\big[N(\kappa)\,(e^J - 1)^+ + N(-\kappa)\,(1 - e^J)^+\big]\,,$$

which reduces to $\frac{S}{2}|e^J - 1|$ *at the money* ($\kappa = 0$)*. Thus expiration reallocates the hedged position away from smooth volatility-level exposure and toward spot-path nonlinearity, at every fixed local moneyness and for both option types. It does not, by maturity alone, distinguish diffusive quadratic variation from discontinuous variation.*

Corollary 1 (Identification consequence)**.** *For ultra-short options, a large Delta-hedging residual is not itself evidence of a jump premium. The same Gamma loading amplifies ordinary discrete-rebalancing error and jump-induced convexity loss. Identification of jump compensation therefore requires a measure that separates continuous from discontinuous spot variation at the hedge horizon, rather than relying on maturity or residual magnitude alone.*

Proposition 2 (Type symmetry of the jump component)**.** *Let* C *and* P *be European call and put prices on the same underlying with the same strike and expiry, produced by any arbitrage-free pricing functional that satisfies put-call parity, and let each position be hedged with the functional's own spot Delta. If the underlying jumps from* S *to* Se^J *at a fixed time while the hedges are held, the hedge residuals of the call and of the put are identical, for every strike, maturity, and* J*.*

Proposition 2 is a two-line parity argument (Appendix B), and it is model-free in the relevant sense: it does not require Black-Scholes, only European parity and hedge ratios that are spot derivatives of the pricing functional. Its force is entirely empirical. The mechanical jump exposure of same-strike calls and puts is one and the same object, so any measured call-put asymmetry in jump-related hedging returns cannot be produced by convexity mechanics; it must originate in the pricing and measure channels of Equation (3), in demand or information structure, or in measurement. Section 5 turns the symmetry into a direct measurement, and Section 7 reads the historical asymmetry in that light.

Remark 1 (Discrete-hedging floor)**.** *Even without jumps or priced state risk, rebalancing at a discrete interval* h *leaves a mechanical residual. For one interval under pure diffusion the hedged P&L is approximately* $\frac{1}{2}\Gamma S^2\sigma^2 h\,(Z^2 - 1)$ *with* Z *standard normal (Boyle and Emanuel, 1980), so that*

$$\mathbb{E}\,|R_{t,h}|\,/S \;\approx\; 2\varphi(1)\,\Gamma S\sigma^2 h, \qquad \mathbb{E}|Z^2 - 1| = 4\varphi(1) \approx 0.968, \tag{4}$$

where φ *is the standard normal density. Equation* (4) *supplies the natural null scale for the same curvature exposure that also loads on small jumps. Equation* (6) *takes that null to the intraday data, where it becomes the unit prediction for* β_C*. The identity* $\mathbb{E}|Z^2 - 1| = 4\varphi(1)$ *is proved in Appendix B.*

Remark 2 (Implied-volatility surfaces)**.** *The Black-Scholes assumption in Proposition 1 is a normalization rather than a constant-volatility restriction. If prices are generated by an implied-volatility surface* $\sigma^{\mathrm{imp}}(K, T)$ *that converges continuously to a positive limit* σ_0 *at* $(S, 0)$*, then along the family* $K = Se^{\kappa\sigma_0\sqrt{T}}$ *the rates in (i) and (ii) and the limits in (iv) hold with* σ_0 *in place of* σ*. Skew moves the constants through* κ*, not the orders in* T*.*

Proposition 1 turns the short-maturity problem into an identification problem. In a calm month, the $T^{-1/2}$ Gamma scale can create a steep maturity gradient with no jump premium at all. On a jump interval, the same curvature exposure amplifies the discontinuity. The empirical task is therefore to establish the diffusive floor first and then ask whether jump measures add explanatory power beyond it.

Three objects recur below and are kept distinct throughout. Jump exposure is the mechanical association between hedged P&L and realized discontinuities, present whenever Gamma is nonzero. Jump-associated excess residual is residual magnitude beyond the estimated continuous hedging floor on identified jump intervals. A jump-risk premium is compensation arising from the wedge

between physical and pricing-measure jump behavior, the object A_{JRP} of Equation (3). The first two are measurable from realized data; only the third is a premium, and neither of the first two establishes it by itself.

3 Modern data and identification

3.1 The intraday panel

The estimation sample is built from the derived option panels of Vilkov (2026), distributed for academic replication at `github.com/vilkovgr/0dte-strategies` under an MIT licence for code. The underlying raw exchange bars are proprietary and are neither used nor redistributed here; the analysis uses only the shipped derived panel, and the replication package for this paper ships code that rebuilds every number from it rather than the panel itself.

The panel records, for each trading date, each thirty-minute mark from 10:00 to 16:00 Eastern, each option type, and a moneyness grid $m = K/S$ running from 0.980 to 1.020 in steps of 0.001, the interpolated mid price and Greeks together with the option's settlement payoff. Four features of the panel were verified before use rather than taken from its documentation, and two of them contradict it.

First, the file contains 1,369,301 rows spanning 2016-09-01 to 2024-05-01, not the approximately 3.5 million rows and September 2016 to January 2026 span stated in the repository's data description. Second, the 16:00 record on each date is not the expiring contract: cross-referencing the package's own volatility panel shows minutes-to-maturity of 1,440 or more at that timestamp, and the at-the-money call premium rises from 7.9 basis points at 15:30 to 33.1 basis points at 16:00. Those records are the next session's contract and are excluded. Third, the intraday records from 10:00 to 15:30 exist only on the 1,380 dates on which a same-day expiry is available; on every other date in the file only the 16:00 record is present. The intraday panel is therefore strictly zero-day-to-expiry, which has a direct consequence for identification recorded in Section 4. Fourth, the shipped column `reth_und` equals exactly $100 \times (\text{payoff} - \text{mid})$, so it is expressed in percent of spot while `mid` and `payoff` are expressed in fractions of spot; the analysis does not use it and reconstructs profit and loss from the price and payoff fields, which satisfy their stated identities exactly.

Units were verified against closed forms rather than assumed. The implied-volatility time convention is calendar time; the ratio of the panel's Gamma to Black-Scholes Gamma has a median of 1.006, the median absolute deviation of the panel's Delta from Black-Scholes Delta is 0.0013, and the panel's mid price matches the Black-Scholes price divided by spot to a median absolute error of 0.13 basis points. The settlement price implied from all twelve entry times agrees to zero basis points within every date. After requiring a positive mid price, positive Gamma, and a finite implied volatility, the estimation sample contains 1,243,969 observations on 1,379 dates, 629,026 calls and 614,943 puts.

3.2 The held-Delta residual

For a contract entered at time u with mid price C_u and hedge ratio Δ_u, held without rebalancing to settlement at 16:00, the residual expressed as a fraction of the entry spot is

$$R_{i,t} = \frac{\Pi_{i,T} - C_{i,u}}{S_u} - \Delta_{i,u}\left(\frac{S_T}{S_u} - 1\right), \tag{5}$$

where $\Pi_{i,T}$ is the settlement payoff. Financing on the hedged position over a horizon of at most six hours is bounded by 0.2 basis points at a five percent rate and is omitted; it is common across contracts within a date and time and is absorbed by the fixed effects below.

Equation (5) is exactly the held-Delta object of Proposition 1, and it is available without interpolation, without a rebalancing convention, and without a pricing model. A single-step held-Delta residual to expiry has a property that shapes the identification: it depends on the terminal spot alone and not on the path. Path quadratic variation therefore has no direct mechanical channel into R. Its influence runs entirely through the distribution of the terminal move, which is why the decomposition below must be read against a simulated benchmark rather than against zero, and why a specification that additionally controls for the realized terminal move isolates a different and narrower object.

The alternative of forming rebalanced thirty-minute returns was considered and rejected. Because the panel indexes contracts by moneyness relative to the contemporaneous spot, tracking a fixed strike across a thirty-minute step requires re-interpolating the grid at the new spot, and the strike leaves the 0.980 to 1.020 grid precisely when the spot moves by more than two percent. That is the selection problem at its worst: contracts would drop out of the panel exactly on the intervals that identify the jump coefficient.

3.3 Continuous and discontinuous variation

Realized variation is measured over each contract's own remaining life $[u, T]$ from the thirty-minute index path, giving $QV_t(u) = \sum_{i \geq u} r_i^2$. The decomposition is $QV = QV^C + JV$, implemented by three estimators.

Intraday volatility is strongly periodic. The median absolute thirty-minute return is 1.387 times its cross-slot average in the 10:00 interval and 1.353 times in the 15:30 interval, against 0.764 at 13:00. Applying a jump test to raw returns therefore concentrates detections mechanically at the open and the close. Returns are standardized by a robust per-slot scale factor normalized to average one before any test is applied, following the periodicity correction of the high-frequency literature; the unadjusted variants are retained as robustness.

The baseline estimator is Lee and Mykland (2008) applied to standardized returns with local volatility from a trailing bipower window of $K = 270$ intraday returns and a Gumbel critical value at the one percent level. It flags 54 of 16,412 tested intervals on 39 distinct dates. The flagged set is dominated by recognizable episodes: the five largest statistics fall on 2017-12-01 at 11:00 and at 11:30, 2019-07-31 at 14:30, 2018-02-05 at 15:30, and 2020-02-26 at 13:30, and the set also contains 2020-02-28, 2020-03-13 at 15:30, and 2021-11-30 at 10:30. The second estimator is threshold truncation, flagging $|r_i|$ above c times the trailing local volatility for $c \in \{3, 4, 5\}$. The third is the bipower shortfall $JV = \max(QV - BV, 0)$ of Barndorff-Nielsen and Shephard (2004) computed over each contract's own life, which requires at least three remaining returns and is therefore reported on the subsample of entries up to 14:30.

3.4 Specification

Under the discrete-hedging benchmark of Remark 1, expected absolute residual is $2\varphi(1)\,\Gamma S\,QV$. Scaling each variance component by that factor gives regressors in basis points,

$$F^C_{i,t} = 2\varphi(1)\,\Gamma_i S_t\,QV^C_t, \qquad F^J_{i,t} = 2\varphi(1)\,\Gamma_i S_t\,JV_t,$$

and the estimating equation is

$$|R_{i,t}| = \alpha + \beta_C F^C_{i,t} + \beta_J F^J_{i,t} + X'_{i,t}\gamma + \mathrm{FE} + \epsilon_{i,t}. \tag{6}$$

Controls are a put indicator, implied volatility, and the quoted bid-ask spread. Fixed effects absorb the twelve entry times and the moneyness grid, with date fixed effects added in an alternative.

Table 1: Continuous versus discontinuous variation in the held-Delta residual

	(1)	(2)	(3)	(4)	(5)
β_C (continuous)	1.151***	0.939***	0.928***	0.313***	0.320***
	(0.058)	(0.079)	(0.067)	(0.043)	(0.050)
β_J (discontinuous)	0.925***	0.833***	0.619***	0.066	-0.218
	(0.095)	(0.104)	(0.079)	(0.070)	(0.151)
λ (realized terminal move)		—		0.508***	0.482***
				(0.028)	(0.027)
$\beta_J - \beta_C$	-0.226	-0.106	-0.309	-0.247	-0.538
p-value	0.051	0.411	0.002	0.003	0.002
p for $\beta_C = 1$	0.009	0.436	0.286	0.000	0.000
Maturity, moneyness FE	No	Yes	Yes	Yes	Yes
Date FE	No	No	Yes	No	Yes
Observations	1,243,969	1,243,969	1,243,969	1,243,969	1,243,969
R^2 (within)	0.532	0.510	0.178	0.669	0.366

Dependent variable is $|R|$ in basis points of index. F^C and F^J are the curvature-scaled continuous and discontinuous components of realized spot variation over each contract's own remaining life, with jumps identified by Lee-Mykland detection on periodicity-standardized thirty-minute returns. Columns (4) and (5) add the realized terminal move scaled by the same curvature factor. Standard errors clustered on 1,379 trading dates in parentheses. Stars denote significance at the ten, five, and one percent levels.

Standard errors are clustered on the trading date, giving 1,379 clusters. Under a pure diffusion with correctly measured QV^C, the benchmark predicts $\beta_C = 1$. The economically relevant object is $\beta_J - \beta_C$, the incremental loading of discontinuous over continuous variation.

The intraday panel contains no contract with a later expiration, so a Jump × 0DTE interaction is not identified and none is estimated. What is available is a within-session maturity clock: time to expiry falls from 360 to 30 minutes across entry times, a twelvefold range. That is a maturity dimension in the sense of Proposition 1, and it is not a cross-expiration comparison; the distinction is maintained in the language throughout.

4 Main results

Table 1 reports the estimates. In the preferred column (2), which absorbs entry time and moneyness, $\beta_C = 0.939$ with a standard error of 0.079. The Boyle-Emanuel prediction of unity cannot be rejected, with a p-value of 0.436. The data are therefore consistent with the curvature channel carrying its theoretical constant, rather than with a curvature channel whose coefficient is merely free to fit.

The discontinuous component adds nothing. Its coefficient is $\beta_J = 0.833$ with a standard error of 0.104, and the contrast $\beta_J - \beta_C = -0.106$ has a p-value of 0.411. Adding date fixed effects in column (3) moves the contrast to -0.309 with a p-value of 0.002, so within a date the jump component loads less than the continuous component rather than more. In no specification is there evidence that identified discontinuities generate excess residual.

Columns (4) and (5) add the realized terminal move under the same curvature scaling. Because the held-Delta residual is a deterministic function of the terminal spot, this control removes the entire mechanical channel, and whatever remains on F^J must operate through the entry price and hedge ratio, that is, through pricing. The jump coefficient falls to 0.066 with a t-statistic of 0.95, and to -0.218 with date fixed effects. The narrower pricing-channel object is also absent.

Table 2: Pure-diffusion placebo calibration of the jump contrast

Replications (four independent seeds)	340
Mean detected jump intervals per replication	46.2
Detected jump intervals, actual data	54
Placebo β_C: mean (sd)	0.858 (0.123)
Placebo β_J: mean (sd)	0.804 (0.348)
Placebo $\beta_J - \beta_C$: mean (sd)	-0.054 (0.357)
standard error of the mean	0.019
range of the four batch means	[-0.120, 0.000]
Placebo $\beta_J - \beta_C$: 5th, 95th percentile	[-0.568, 0.579]
Estimated $\beta_J - \beta_C$	-0.106
Share of replications below the estimate	0.476
Standardised distance from placebo mean	-0.15
Power against a true excess loading	
+0.20	0.09
+0.40	0.20
+0.60	0.39
+0.80	0.61
Minimum excess detectable with 80% power	1.00

Each replication simulates twelve Gaussian thirty-minute returns per date with a jump-robust bipower variance profile and the empirical intraday periodicity, prices options with the same variance that generates the path, and reruns the full detection and estimation pipeline. There is no jump and no risk premium by construction, so the recovered contrast measures detection selection alone. The 340 replications are pooled from four batches with independent seeds; the batch means are reported because they show how unstable the location of this distribution is at small replication counts. Power is computed against the pooled null standard deviation at a two-sided five percent level.

4.1 Calibrating the null

A contrast that is not statistically distinguishable from zero is uninformative unless the distribution it should be compared against is known. Table 2 supplies it. Across 340 replications of a jump-free, premium-free world, the detection pipeline flags 46.2 intervals per replication against 54 in the data, and recovers β_C with a mean of 0.858 and $\beta_J - \beta_C$ with a mean of -0.054 and a standard deviation of 0.357. The estimated contrast of -0.106 lies 0.15 standard deviations below the placebo mean, and 48 percent of replications fall below it.

The replication count matters more than it might appear. Pooled across four independent batches the mean is estimated to within 0.019, but the individual batch means range from -0.120 to 0.000, a spread comparable to the empirical contrast itself. A placebo run at the twenty replications of a smaller exercise would locate this distribution only to within roughly its own width, so the number of replications is reported alongside the moments.

Two conclusions follow, and the second is a limitation. The estimate is what a world without jumps produces, which is the strongest available form of the null. But the null distribution is wide: its standard deviation of 0.357 is nearly three times the analytic date-clustered standard error of 0.129, and the power calculation in Table 2 shows that a true excess of +0.60 would be detected only 39 percent of the time. An excess of about one times the continuous loading is the smallest this design detects with eighty percent power. The result is therefore a well-identified null, not a tight bound, and the earlier reading under which the design excluded an excess of 0.6 or larger is not supported by the expanded placebo.

The placebo also calibrates the level. A one-step held-Delta position to expiry is not the

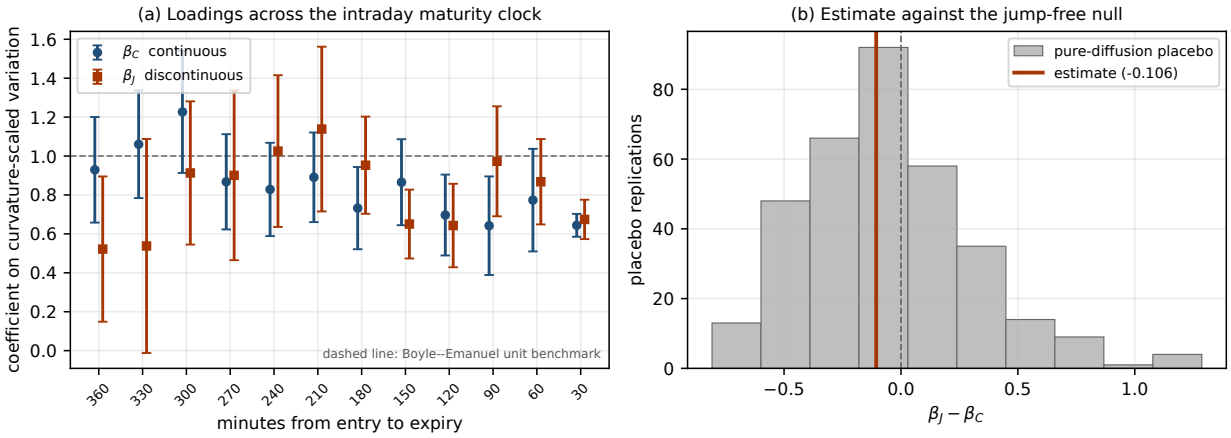


Figure 2: Panel (a) plots β_C and β_J estimated separately at each entry time, with ninety-five percent date-clustered intervals; the dashed line is the Boyle-Emanuel unit benchmark. Panel (b) plots the distribution of $\beta_J - \beta_C$ across pure-diffusion placebo replications against the estimate.

Table 3: Inference on the contrast $\beta_J - \beta_C$

Inference procedure	$\beta_J - \beta_C$	se or sd	p
Date-clustered analytic (baseline)	-0.106	0.129	0.411
Wild cluster bootstrap, 599 reps	-0.106	—	0.392
Date randomisation, 60 permutations	-0.106	0.372	0.867
Aggregated date × entry time (16,548 cells)	-0.088	0.155	0.569
Pure-diffusion placebo, 340 replications	-0.106	0.357	—
Leave-one-jump-date-out, 39 dates			
range of the contrast	[-0.202, -0.053]		
specifications significant at five percent	0 of 39		

All procedures apply to column (2) of Table 1. The wild cluster bootstrap draws Rademacher weights on trading dates with the restriction $\beta_J = \beta_C$ imposed. The randomisation test permutes each date's jump share across dates while holding total curvature-scaled variation fixed, so the null it imposes is that the flagged dates are not special rather than that discontinuities do not exist. The aggregated specification collapses the panel to date by entry-time cell means. The leave-one-out exercise drops each of the 39 dates carrying jump variation in turn.

small-step limit for which the Boyle-Emanuel quadratic expansion is exact, and the placebo recovers $\beta_C = 0.858$ with a standard deviation of 0.123 rather than exactly one. The estimated 0.939 lies inside that distribution, so the empirical loading is consistent both with the analytical benchmark and with its finite-step simulated counterpart.

4.2 Inference and the effective identifying sample

The panel has 1,243,969 rows, but the number of rows is not the number of observations that identify β_J. Lee-Mykland flags 54 intervals on 39 dates, and only 28,053 rows, 2.3 percent of the panel, carry positive jump variation. Date-clustered standard errors are the right first step, and their asymptotics run in the number of clusters that carry identifying variation rather than in the 1,379 clusters that exist. Table 3 therefore reports five procedures side by side rather than selecting one.

The five agree. The bootstrap p-value of 0.392 is close to the analytic 0.411, so the cluster-robust approximation is not doing damage here. The aggregated specification, which discards all within-cell variation and estimates on 16,548 cells, gives -0.088 with a p-value of 0.569. The randomisation

test is the most informative: reassigning which dates count as jump dates produces a contrast at least as large in absolute value 87 percent of the time, so the flagged dates are not distinguishable from arbitrary dates carrying the same total variation. No single date drives the result. Dropping the March 2020 crash moves the contrast to -0.202 and dropping February 2018 moves it to -0.173, both still insignificant, and none of the 39 leave-one-out estimates is significant at five percent.

Two of these procedures also bear on precision. The randomisation standard deviation of 0.372 and the placebo standard deviation of 0.357 both sit near three times the analytic clustered standard error of 0.129, so the analytic error understates the variability of this contrast under reassignment of the identifying events. The power calculation in Table 2 uses the simulated scale for that reason.

4.3 The maturity clock

Mean absolute residual rises monotonically across the session, from 2.00 basis points at thirty minutes to expiry to 9.16 basis points at six hours. Interacting both components with log horizon gives a continuous interaction of 0.198 with a t-statistic of 4.48 and a jump interaction of 0.050 with a t-statistic of 0.67: the curvature loading strengthens as horizon lengthens, and the jump loading does not vary with horizon. Panel (a) of Figure 2 estimates Equation (6) separately at each of the twelve entry times. The continuous loading is near unity at long intraday horizons and falls toward 0.64 in the final half hour, consistent with the quadratic expansion degrading as the payoff kink dominates. The jump loading is statistically indistinguishable from the continuous loading at the five percent level at every one of the twelve horizons. Three contrasts are marginal at ten percent: $+0.331$ at ninety minutes, -0.523 at five and a half hours, and -0.407 at six hours. Two of those three point away from a jump excess and one toward it, and three marginal rejections in twelve tests is within what multiplicity alone produces; no horizon supports a jump excess at conventional levels.

5 Calls versus puts

Proposition 2 states that for matched European calls and puts hedged with parity-consistent Deltas the jump component of the hedged residual is identical. The intraday panel places both types on the same strike grid at the same timestamps, so the proposition is testable directly. Under exact parity $R_{\text{call}} - R_{\text{put}}$ vanishes identically, so any measured difference is a measurement of departures from parity in quoted mid prices and vendor hedge ratios, not a convexity effect.

Table 4 reports the result on 565,573 matched pairs. The mean residual difference is 0.055 basis points and its mean absolute value is 1.076 basis points, against a mean absolute residual of 5.608 basis points in the panel. The proposition therefore accounts for roughly eighty percent of the residual magnitude as common across types. The remaining difference is fully consistent with measured parity violations in the inputs: the mean difference of the two vendor Deltas is 0.9904 where parity requires exactly one, and the mean absolute call-put price parity gap is 0.864 basis points.

The mean difference is statistically distinguishable from zero, with a date-clustered t-statistic of 2.70. With 1,379 clusters and more than half a million pairs, that is unsurprising and is not economically meaningful: 0.055 basis points is one percent of the typical residual. The correct reading is that parity holds to within the measurement error of the inputs, not that it holds exactly.

The parity-sharp interaction is the direct implementation of the test set out in Section 3. With $|R|$ as dependent variable, $\beta_P = 0.0088$ with a standard error of 0.0083 and a p-value of 0.289, consistent with the proposition. With signed R, $\beta_P = 0.0164$ with a p-value of 0.028. Panel (b) of Figure 3 shows why the distinction matters: the mean absolute residual quadruples across the session while the mean absolute call-put difference stays flat near one basis point.

Table 4: Parity test on matched calls and puts

Matched call–put pairs (same date, entry time, strike)	565,573
Mean $R_{\text{call}} - R_{\text{put}}$ (bp)	0.055
Median (bp)	-0.037
Mean absolute difference (bp)	1.076
Standard deviation (bp)	1.905
Mean $\lvert R \rvert$ in the panel (bp)	5.608
Date-clustered t on the mean difference	2.70
Mean $\Delta_{\text{call}} - \Delta_{\text{put}}$ (parity value 1)	0.9904
Mean absolute call–put price parity gap (bp)	0.864
β_P: jump exposure × put, dependent variable $\lvert R \rvert$	0.0088
standard error / p	(0.0083) / 0.289
β_P: jump exposure × put, dependent variable R	0.0164
standard error / p	(0.0075) / 0.028

Pairs are matched on date, entry time, and moneyness. β_P is the coefficient on the interaction of curvature-scaled jump variation with a put indicator, estimated with entry-time and moneyness fixed effects and date-clustered standard errors.

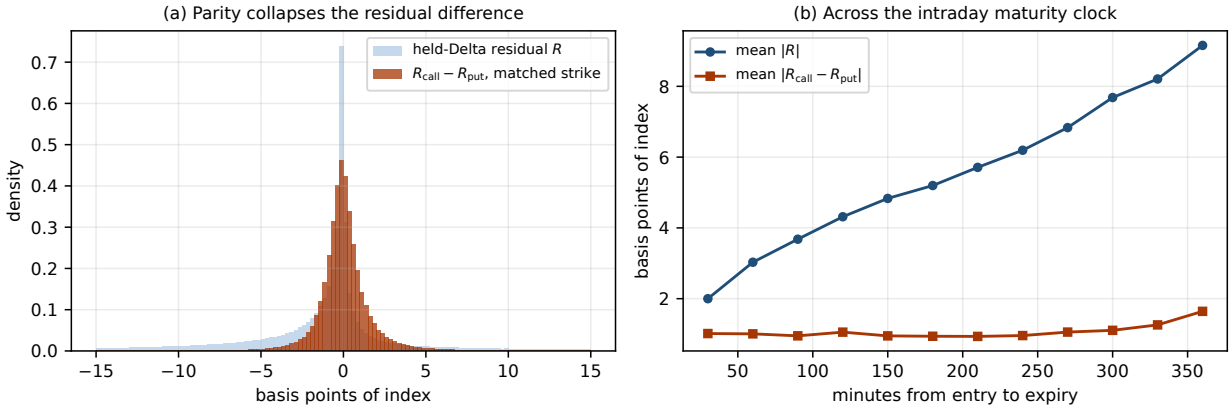


Figure 3: Panel (a) compares the distribution of the held-Delta residual with the distribution of the matched-strike call-put difference. Panel (b) plots both magnitudes across the intraday maturity clock.

The signed-residual interaction is small but statistically present, and the paper does not read it as evidence of demand or information transmission. Proposition 2 rules out the convexity channel; it does not adjudicate among the remaining ones. Pricing, liquidity, contract composition, microstructure, hedge measurement, and information are all compatible with an effect of this magnitude, and the measured parity violations in the inputs are by themselves of the right order to produce it.

6 Robustness

Table 5 collects the robustness evidence. Across nine jump definitions, β_C lies between 0.875 and 1.180 and β_J between 0.833 and 1.042. The contrast is statistically insignificant in eight of the nine. The exception is threshold truncation at $c = 3$, the loosest rule, which flags 606 intervals against 54 under the baseline and gives a contrast of -0.209 with a p-value of 0.032; it points in the same direction as every other specification, away from a jump excess. Replacing the thirty-minute total

Table 5: Jump definitions, hedge ratios, and volatility regimes

Jump definition	β_C	β_J	$\beta_J - \beta_C$	p
Lee–Mykland, $K = 270$ (baseline)	0.939 (0.079)	0.833 (0.104)	-0.106	0.411
Lee–Mykland, $K = 156$	0.875 (0.079)	0.910 (0.091)	+0.034	0.776
Lee–Mykland, $K = 390$	0.927 (0.086)	0.849 (0.095)	-0.078	0.546
Lee–Mykland, no periodicity adjustment	0.917 (0.089)	0.861 (0.086)	-0.056	0.645
Threshold truncation, $c = 3$	1.067 (0.082)	0.858 (0.070)	-0.209	0.032
Threshold truncation, $c = 4$	0.971 (0.101)	0.854 (0.076)	-0.117	0.324
Threshold truncation, $c = 5$	0.975 (0.094)	0.833 (0.086)	-0.141	0.253
Bipower shortfall (BNS)	0.888 (0.107)	0.856 (0.121)	-0.032	0.863
Baseline, one-minute total QV	1.180 (0.089)	1.042 (0.085)	-0.138	0.264
Hedge ratio				
Vendor implied-volatility Delta	0.939 (0.079)	0.833 (0.104)	-0.106	0.411
Black–Scholes Delta from quoted IV	0.934 (0.078)	0.839 (0.102)	-0.095	0.455
Sticky-moneyness (smile-adjusted) Delta	0.940 (0.075)	0.764 (0.110)	-0.176	0.157
Minimum-variance Delta, leave-one-year-out	0.940 (0.073)	0.866 (0.104)	-0.074	0.549
Volatility regime (date-level implied-volatility terciles)				
Low	0.879 (0.081)	0.076 (0.127)	-0.803	0.000
Middle	0.961 (0.100)	0.939 (0.480)	-0.023	0.964
High	0.865 (0.103)	0.815 (0.112)	-0.051	0.728

Each row re-estimates column (2) of Table 1. Bipower shortfall requires at least three remaining thirty-minute returns and is estimated on 1,044,403 observations. The minimum-variance Delta is of Hull-White form and is estimated leave-one-year-out, so it is never fitted on the observations where it is evaluated.

variation with the one-minute realized variance shipped in the source package raises both coefficients proportionately and leaves the contrast at -0.138.

Four hedge conventions are compared. The vendor implied-volatility Delta, a Black-Scholes Delta recomputed from the quoted implied volatility, a sticky-moneyness Delta built from the slope of the implied-volatility surface across the moneyness grid, and a minimum-variance Delta of Hull and White (2017) form. The minimum-variance coefficients are estimated leave-one-year-out and applied to the held-out year, so no in-sample fit is reported as robustness. Mean absolute residual is 5.608 basis points under the vendor Delta, 5.569 under the recomputed Black-Scholes Delta, 5.428 under sticky moneyness, and 5.596 under the out-of-sample minimum-variance Delta. The fitted hedge improves on the vendor Delta by 0.012 basis points, two tenths of one percent, which is no material improvement; the largest reduction, three percent, comes from the sticky-moneyness rule rather than from the fitted one. Both facts are reported here rather than suppressed. Under all four conventions β_C is between 0.934 and 0.940 and the contrast remains insignificant. The absence of a fitted-hedge gain is itself consistent with a literature that has found flexible hedge ratios hard to convert into out-of-sample gains: Buraschi and Jackwerth (2001) show that hedging performance is informative about which factors options span, while Qiao and Wan (2024), Zernikov (2026), Wysocki and Ślepaczuk (2024), and Brini et al. (2025) report that learned or heavily parameterized hedges recover much of their advantage from the same curvature and smile structure that the closed-form Deltas already contain.

Splitting dates into terciles of average implied volatility, the contrast is -0.803 with a p-value below 0.001 in the low-volatility tercile and statistically indistinguishable from zero in the middle and high terciles. The direction is again away from a jump excess, and it is the calm regime rather than the stressed one in which discontinuous variation loads least.

Taken together, the evidence is not that every specification returns the same insignificant contrast. The accurate summary is that no specification provides evidence of a positive excess

loading on discontinuous variation, while some specifications estimate a significantly smaller loading for discontinuous than for continuous variation: date fixed effects give -0.309 with a p-value of 0.002, the loosest truncation rule gives -0.209 at 0.032, and the calm tercile gives -0.803 below 0.001. Those three are the specifications that condition most tightly on the days and returns where detection fires, and a negative contrast is what selection predicts once the continuous component absorbs the curvature scale. None of them can be read as a jump premium of the wrong sign, because the placebo produces contrasts of comparable magnitude in a world with no jumps at all. What they do rule out is the reading under which identified discontinuities carry an excess loading that the pooled specification is merely too imprecise to detect.

No scheduled-announcement calendar is assumed. The intraday panel carries no event indicator of its own, and no announcement calendar aligned to its timestamps was available under terms that permit use here, so no event cut is reported; the replication code accepts a user-supplied calendar as an input for researchers who hold one. The volatility-regime split above is the state-dependence evidence the data support without an external calendar.

7 Prior evidence and the daily-frequency diagnostic

The question this paper takes to intraday data was raised by the delta-hedged gains literature, which uses hedged option returns to isolate priced risk that a local hedge does not remove (Bakshi and Kapadia, 2003). Within that literature the call and put sides of the same strike have not always behaved alike, and asymmetries of that kind have been attributed to demand pressure, to deviations from parity in quoted prices, and to measurement (Gârleanu et al., 2009; Cremers and Weinbaum, 2010). Proposition 2 constrains that discussion from the theory side. Because the mechanical jump exposure of same-strike calls and puts is one object under parity, and because Section 5 confirms that identity in modern data to within one basis point out of five and a half, no measured call-put asymmetry in jump-related hedging returns can be a convexity artifact at any maturity. It must originate in the pricing and measure channels of Equation (3), in demand or information structure, or in measurement. That is a restriction on the candidate explanations, not a test among them.

A February 2023 public WRDS/OptionMetrics call extract of 2,070 matched one-day observations supports the daily-frequency diagnostic that motivated the intraday design. Mean absolute normalized residual is 9.88 basis points at one to seven days against 5.96 basis points at thirty-one to sixty days; with trading dates as paired units the equal-date gap is 3.09 basis points, positive on 14 of 18 dates, with an exact sign-flip reference p-value of 0.005 under sign symmetry. Netting each contract's own discrete-diffusion floor removes that gradient and reverses it, to -2.72 basis points with a sign-flip p-value of 0.019, and the removal of the positive gradient survives floor calibrations built from trailing realized and bipower variation, although the significance of the reversal is specific to the implied calibration. That daily result is what the intraday experiment was designed to adjudicate, and Section 4 adjudicates it: the gradient is a curvature phenomenon, and once curvature is scaled correctly there is no discontinuity residual left above it.

8 Conclusion

On more than 1.2 million intraday observations of SPXW zero-day-to-expiry options, curvature-scaled continuous quadratic variation accounts for the held-Delta residual with a loading statistically indistinguishable from the discrete-diffusion benchmark of one, and identified discontinuous variation adds nothing above it. No specification provides evidence of a positive excess loading on discontinuous variation, and several estimate a significantly smaller loading for discontinuous than for continuous

variation. The result survives the identification concern that motivates the design, because the estimated contrast is what a simulated jump-free world produces under the same detection procedure. Matched calls and puts share their residual to within one basis point out of five and a half, confirming the parity proposition directly and removing convexity from the list of candidate explanations for any measured call-put asymmetry in jump-related hedging returns.

The finding is a null on jump excess and a positive result on what curvature explains: in the ultra-short segment a large Delta-hedging residual is a Gamma phenomenon, carried by the continuous channel at its own theoretical constant.

A local Delta hedge removes first-order spot exposure, but the residual near expiration is dominated by curvature. Once the continuous curvature-driven hedging floor is properly accounted for, observed discontinuities in modern SPXW 0DTE data do not generate the large additional residual contribution that a naive jump-event comparison might suggest. That conclusion concerns realized hedging residuals. It is not an identification of the jump-risk premium: the residual is measured under the physical measure against a pricing input taken as given, and no wedge between physical and pricing-measure jump behavior is estimated. Identifying one would require either a maturity-spanning intraday panel, which the source data do not contain, or an event calendar aligned to the intraday timestamps.

Data provenance and reproducibility

The intraday results rest on the derived option panels of Vilkov (2026), obtained from the public repository accompanying that paper and verified by SHA-256 against the distributed object hashes before use. Those panels are provided by their author for academic replication and are not redistributed with this paper, and neither are the proprietary exchange bars from which they were built. The replication package ships the code that rebuilds every intraday number from them, the derived date-level variation and jump series, all estimation output, the placebo and inference modules, and both figures; a reader who obtains the panels from their source can reproduce Sections 4 to 6 in full, and can audit jump identification from the distributed series alone. No estimate reported in this paper is taken from another source; every coefficient is produced by the code shipped here. The February 2023 diagnostic regenerates from the matched panel distributed with this package and reproduces bit-identically; the raw vendor extract behind it is licensed and is not included.

A Fuller residual-risk decomposition

Applying Ito's lemma with jumps to $C(t, X_t)$, imposing the corresponding risk-neutral pricing restriction, and substituting the result into the Delta-neutral portfolio removes the first-order continuous spot term but leaves the non-spot diffusion premium, random diffusion shocks to non-spot states, jumps in those states, the spot jump compensator, and the difference between physical- and pricing-measure jump behavior. In the notation of Section 2, a schematic version of the realized residual is

$$R_{t,h} = \int C_q \lambda_q \, du + \int C_q \Sigma_q \, dW_q + \int C_q \, dJ_q + \int C_S S \, m_J^S \, du + \int (\text{jump-measure wedge}) \, du. \quad \text{(A1)}$$

For sufficiently short h, the conditional expectation removes the mean-zero diffusion innovation and yields Equation (3) in the main text. The empirical design does not require every theoretical channel to be separately nonzero. Its test is whether observable jump measures retain incremental explanatory power for the hedged return after the controls absorb the most direct volatility, model-risk, contract, and information-transmission effects.

B Proofs for Section 2.2

Proof of Proposition 1. Fix κ and set $K = Se^{\kappa\sigma\sqrt{T}}$, so that

$$d_1 = \frac{\ln(S/K) + (r+\sigma^2/2)T}{\sigma\sqrt{T}} = -\kappa + \frac{(r+\sigma^2/2)\sqrt{T}}{\sigma} \longrightarrow -\kappa, \qquad d_2 = d_1 - \sigma\sqrt{T} \longrightarrow -\kappa.$$

(i) Vega is $S\varphi(d_1)\sqrt{T}$ for the call and the put alike, so Vega $\sim S\varphi(\kappa)\sqrt{T}$, and the convergence is uniform for κ in compact sets because φ is bounded.

(ii) Gamma is $\varphi(d_1)/(S\sigma\sqrt{T})$ for both types, so $\Gamma \sim \varphi(\kappa)/(S\sigma\sqrt{T})$, with the same uniformity.

(iii) Hold the initial Delta fixed over a small spot move $\Delta S = Su$. A second-order Taylor expansion of either pricing function gives

$$V(S+\Delta S, T) - V(S,T) - \Delta\,\Delta S = \tfrac{1}{2}\Gamma(\Delta S)^2 + o((\Delta S)^2) = \tfrac{1}{2}\Gamma S^2 u^2 + o(u^2),$$

with the same Γ for the call and the put. The expansion depends on the realized spot increment, not on whether that increment was generated by the continuous or jump component of the semimartingale. Thus small continuous moves and small jumps share the same local curvature loading.

(iv) For a fixed log jump J, the call residual is $C(Se^J, T) - C(S,T) - N(d_1)S(e^J - 1)$. As $T \to 0$, $K \to S$, so $C(Se^J, T) \to (Se^J - S)^+ = S(e^J - 1)^+$, $C(S,T) \to (S-K)^+ \to 0$, and $N(d_1) \to N(-\kappa)$. The limit is $S[(e^J-1)^+ - N(-\kappa)(e^J-1)]$, which equals $S\,N(\kappa)(e^J-1)$ for $J > 0$ and $S\,N(-\kappa)(1-e^J)$ for $J < 0$, that is,

$$S\big[N(\kappa)\,(e^J-1)^+ + N(-\kappa)\,(1-e^J)^+\big] > 0 \quad \text{for } J \neq 0.$$

The put residual is $P(Se^J, T) - P(S,T) - (N(d_1)-1)S(e^J-1) \to S(1-e^J)^+ + N(\kappa)S(e^J-1)$, and the same case split yields the identical expression. At $\kappa = 0$ both reduce to $\frac{S}{2}|e^J - 1|$. Parts (i)-(iv) establish that smooth volatility-level sensitivity vanishes while spot-path nonlinearity becomes increasingly important, but they do not identify whether realized quadratic variation is continuous or discontinuous. □

Proof of Proposition 2. European put-call parity states $C(S,\cdot) - P(S,\cdot) = Se^{-qT} - Ke^{-rT}$ at every spot level (with Se^{-qT} replaced by S less the present value of discrete dividends where applicable), for any arbitrage-free pricing functional. A jump from S to Se^J at a fixed time therefore changes $C - P$ by $e^{-qT}S(e^J - 1)$. Differentiating parity in S gives $\Delta_C - \Delta_P = e^{-qT}$ for hedge ratios that are spot derivatives of the pricing functional. The difference of the two hedge residuals is

$$\big[C(Se^J) - C(S)\big] - \big[P(Se^J) - P(S)\big] - (\Delta_C - \Delta_P)\,S(e^J - 1) = e^{-qT}S(e^J-1) - e^{-qT}S(e^J-1) = 0. \square$$

Note on Remark 2. If prices are $V(S,K,T) = V_{\mathrm{BS}}(S,K,T,\sigma^{\mathrm{imp}}(K,T))$ with $\sigma^{\mathrm{imp}}(K,T) \to \sigma_0 > 0$ continuously at $(S,0)$, then along $K = Se^{\kappa\sigma_0\sqrt{T}}$ one has $d_1 \to -\kappa\sigma_0/\sigma_0 = -\kappa$, and substituting σ^{imp} into the closed forms above reproduces the rates and limits of Proposition 1 with σ_0 in place of σ.

The benchmark constant. For Z standard normal,

$$\mathbb{E}|Z^2 - 1| = 2\int_0^1 (1-z^2)\varphi(z)\,dz + 2\int_1^\infty (z^2-1)\varphi(z)\,dz.$$

Integration by parts gives $\int_a^\infty z^2\varphi(z)\,dz = a\varphi(a) + 1 - N(a)$, hence $\int_1^\infty (z^2-1)\varphi\,dz = \varphi(1)$, and $\int_0^1 z^2\varphi\,dz = N(1) - \frac{1}{2} - \varphi(1)$, hence $\int_0^1 (1-z^2)\varphi\,dz = \varphi(1)$. Therefore $\mathbb{E}|Z^2-1| = 4\varphi(1) \approx 0.9679$. Substituting into the one-interval expansion $R \approx \frac{1}{2}\Gamma S^2\sigma^2 h(Z^2-1)$ of Boyle and Emanuel (1980) yields Equation (4). □

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